

InvoiceIQ

Invoice Key-Field Extraction from Images and PDFs

Client overview · Prepared for sharing · Raunak Rath

1. Executive Summary

InvoiceIQ extracts structured business fields from invoice images or PDFs — vendor, dates, totals, payment terms, and line items — without hand-built templates. It tries a vision model when available, then OCR, and falls back to a reliable demo path for presentations.

Primary users: AP automation, finance ops, document-AI product demos.

2. Key Features

- Upload PNG, JPG, TIFF, or PDF invoices
- Extract header and financial fields with confidence scores
- Line-item table extraction
- Dashboard of extraction volume and amounts
- History of past extractions
- Upcoming-feature roadmap in the UI

3. How the System Works (Code Flow)

The application follows a simple layered design: React dashboard → FastAPI backend → core processing module → SQLite storage.

1. User uploads an invoice file from the dashboard.
2. Backend saves the file; PDF first page is converted to an image.
3. Extractor tries vision-language model JSON extraction first.
4. If unavailable, OCR (Florence-2 / Tesseract) + field parsing is used.
5. In demo mode, a structured schema is returned so demos never fail.
6. Fields and line items are normalized and stored in SQLite.
7. UI shows a preview, field grid, totals, and history.

4. Technology Stack

- Frontend: React + Vite + Tailwind (port 5173)
- Backend: FastAPI + SQLite + Pillow + PyMuPDF (port 8003)
- AI: Ollama vision / Florence-2 / Tesseract; demo mode without VLM

5. How to Run

```
cd invoiceiq/backend && DEMO_MODE=1 uvicorn main:app --reload --port 8003
cd invoiceiq/frontend && npm install && npm run dev
Open http://127.0.0.1:5173
```

6. Sample Files for Demo

- samples/sample_invoice.png — recommended demo image
- samples/sample_invoice.jpg — same invoice as JPEG
- samples/sample_invoice.pdf — same invoice as PDF
- Sample vendor: Acme Cloud Labs · Invoice INV-48291

7. Main API Endpoints

- POST /api/extract — extract invoice fields
- GET /api/extractions — list past runs
- GET /api/extraction/{id} — fetch one result
- GET /api/stats — dashboard metrics

8. Upcoming Features

The following capabilities are planned for future releases and are shown in the product UI under Upcoming.

- Multi-page PDF extraction
- Human correction and CSV/JSON export
- Stronger line-item table parsing
- Confidence calibration for noisy OCR

9. Folder Layout

- invoiceiq/frontend — web dashboard
- invoiceiq/backend — API and core logic
- invoiceiq/samples — demo upload / paste files
- invoiceiq/docs — internal report and slides